

Influences of ENSO, NAO, IOD and PDO on seasonal precipitation regimes in the Yangtze River basin, China

Mingzhong Xiao,^{a,b,c} Qiang Zhang^{a,b,c*} and Vijay P. Singh^{d,e}

^a Department of Water Resources and Environment, Sun Yat-sen University, Guangzhou, China

^b Key Laboratory of Water Cycle and Water Security in Southern China of Guangdong High Education Institute, Sun Yat-sen University, Guangzhou, China

^c School of Geography and Planning, and Guangdong Key Laboratory for Urbanization and Geo-simulation, Sun Yat-sen University, Guangzhou, China

^d Department of Biological and Agricultural Engineering, Texas A & M University, College Station, USA

^e Department of Civil and Environmental Engineering, Texas A & M University, College Station, USA

ABSTRACT: Teleconnections between El Niño/Southern Oscillation (ENSO), North Atlantic Oscillation (NAO), Indian Ocean Dipole (IOD) and Pacific Decadal Oscillation (PDO) and seasonal precipitation regimes over the Yangtze River basin have been analysed based on the rotated empirical orthogonal functions. Results show that ENSO is the leading driver of seasonal precipitation variability over the Yangtze River basin, and the spring precipitation has been influenced by the PDO and ENSO, the summer and autumn precipitation has been influenced by the ENSO and IOD, the winter precipitation has been influenced by the ENSO, IOD and NAO. Furthermore, changes for the seasonal occurrence and intensity of wet days linked to the ENSO, NAO, IOD and PDO indices have also been investigated to discover which is the dominant mechanism driving seasonal precipitation changes. And results indicated that the influences of ENSO, NAO, IOD and PDO on the seasonal occurrence and intensity of precipitation events are complex, such as that the negative PDO event at the same year tends to increase the spring occurrence of precipitation events in the southwestern part of the Yangtze River basin while the positive ENSO event a year earlier tends to increase the spring intensity of precipitation events in the east part of the Yangtze River basin.

KEY WORDS ENSO; precipitation; the Yangtze River basin; rotated empirical orthogonal functions; the occurrence of wet days; the intensity of wet days

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1. Introduction

Variations of precipitation regimes in both space and time can significantly influence the spatiotemporal distributions of water resources and can result in either severe floods or droughts, which exercise a definitive influence on society and regional economies (Kripalani and Kulkarni, 2001). And this variability is often linked to both internally and externally forced fluctuations in the global sea surface temperature (SST) field (Pui *et al.*, 2012). El Niño/Southern Oscillation (ENSO) is a coupled ocean–atmosphere tropical Pacific phenomenon with global reach. On inter-annual timescales the significant part of the global climatic variation can be linked to ENSO (Webster *et al.*, 1998). In recent years, the possible influences of ENSO events on precipitation regimes in both space and time have been receiving increasing attention, and many researchers have found significant influences of ENSO events on precipitation events at different places, e.g. the United States

(Gershunov and Barnett, 1998), North Pacific (Gershunov and Cayan, 2003), South America (Grimm and Tedeschi, 2009) and China (Liu and Ding, 1995; Gong and Wang, 1999; Zhang *et al.*, 1999; Lan *et al.*, 2002; Li *et al.*, 2005).

The Yangtze River (Figure 1), being the longest river in China and the third longest river in the world, plays an important role in the sustainable development of economy and ecology in China. It is found that disastrous floods that occurred in the entire Yangtze River basin in 1998 was a direct result of unusually high precipitation (670 mm) that occurred between June and August due to a strong El Niño event (Yin and Li, 2001). Since then, many investigations on the floods and possible mechanisms have been undertaken. Jiang *et al.* (2006) analysed the teleconnections between flood/drought events in the Yangtze River basin and ENSO events during 1868–2003 with the help of χ^2 test and spectral analysis, and it is found that in the lower and middle Yangtze catchments, El Niño shows a close relation with flood events and La Niña correlates with drought events. However, the relationships are reversed for the upper Yangtze catchment. Zhang *et al.* (2007) analysed the possible teleconnections between ENSO and annual maximum streamflow from the

* Correspondence to: Q. Zhang, Department of Water Resources and Environment, Sun Yat-Sen University, Guangzhou 510275, China. E-mail: zhangq68@mail.sysu.edu.cn

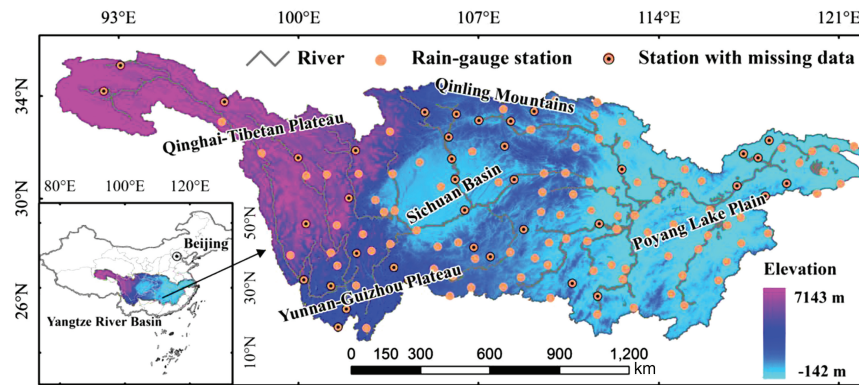


Figure 1. Locations of the Yangtze River basin and the rain-gauge stations.

lower, the middle and the upper Yangtze River basin using the cross-wavelet analysis, and results indicated that in the lower Yangtze Basin, the in-phase relationship occurred between annual maximum streamflow and ENSO. However, in the upper Yangtze Basin, the anti-phase relationship is dominant, and ambiguous in the middle Yangtze. Besides, to predict streamflow one season ahead, the climatic influences on summer monsoon inflow to the Three Gorges Dam in the Yangtze River basin have also been analysed by Xu *et al.* (2007). The streamflow is influenced by the precipitation, and then it is also necessary to analyse the possible teleconnection between the climate indices and seasonal precipitation in the Yangtze River basin. However, few reports have been available so far on this subject, especially those focusing on the Yangtze River basin.

Meanwhile, there are many articles analysing the summer monsoon in the Yangtze River basin and the associated large scale circulation, it has been well studied that the East Asian Monsoon has been significantly influenced by the ENSO (Chen *et al.*, 2013). Besides, the East Asian Monsoon has also been influenced by the North Atlantic Oscillation (NAO), Indian Ocean Dipole (IOD) and Pacific Decadal Oscillation (PDO) (Yuan *et al.*, 2008; Linderholm *et al.*, 2011; Chen *et al.*, 2013). Linderholm *et al.* (2011) found that the major patterns of summer climate over China are highly connected with the interannual variation of the summer NAO, supporting a teleconnection between the North Atlantic region and East Asia. And the influences of ENSO on the East Asian Monsoon are always modulated by the IOD and PDO (Chakravorty *et al.*, 2013; Chen *et al.*, 2013). Then it is necessary to analyse the possible influences of ENSO, NAO, IOD and PDO on the seasonal precipitation in the Yangtze River basin, which has not been studied.

With both daily and sub-daily precipitation data in east Australia, Pui *et al.* (2012) found that it is the occurrence of rainfall events, rather than the average intensity of rainfall during the events that is the dominant mechanism driving seasonal rainfall changes caused by ENSO, IOD and the southern annular mode. Then also to discover the dominant mechanism driving seasonal precipitation changes caused by ENSO, NAO, IOD and PDO, changes for the number

and intensity of wet days (daily precipitation ≥ 1 mm) in each season linked to the ENSO, NAO, IOD and PDO indices has been investigated in this article.

The objectives of this article are therefore to investigate (1) the influences of ENSO, NAO, IOD and PDO on the seasonal precipitation regimes across the Yangtze River basin; and (2) whether ENSO, NAO, IOD and PDO are the dominant mechanism behind the changes of occurrence or intensity of precipitation events across the Yangtze River basin. This study will provide valuable information for improving the long-term forecasting of precipitation using its relationship with ENSO, NAO, IOD and PDO.

2. Study region and datasets

2.1. Study region

The Yangtze River, with a drainage area of 1.8×10^6 km², is the longest river in China and the third longest river in the world, ranking after the Nile River in Africa and Amazon River in South America. Originating from the Qinghai-Tibetan Plateau at an elevation higher than 5000 m, the Yangtze River flows across three major morphological steps from the western plateau to the east coast, a 6000 m difference in elevation (Chen and Zhao, 2001). The river basin is located in the subtropical and temperate climate zone and the climate of the region are dominated by the Monsoon winds. In climate, the southern part of the Yangtze River basin is climatically close to subtropical climate and northern part to the temperate zone. The annual mean temperature in the southern and northern parts of the middle and lower Yangtze River basin is 19 and 15 °C, respectively; and the mean annual precipitation in the basin varies from 300 to 500 mm in the western region to 1600–1900 mm in the southeastern region and precipitation is mostly concentrated in the summer season (from June to August), accounting for 70–80% of the annual total precipitation amount (Yu *et al.*, 2009).

2.2. Datasets

Daily precipitation data covering a period of 1960–2009 were collected at 135 rain stations which are provided

by the National Meteorological Information Center of the China Meteorological Administration (Figure 1). There are 38 rain gauge stations containing missing daily precipitation data (Figure 1). However, only one station had 1.09% missing values, and most of them had less than 0.1% of total missing values. These series with missing values were replaced by the long-term average of the same days of other years.

In the study the ENSO are indicated by Niño 3.4 indices, derived from SST anomaly estimates in the Niño 3.4 region (5°N–5°S, 120–170°W). And the Niño 3.4 indices were extracted from the Climate Prediction Center of National Oceanic and Atmospheric Administration (NOAA) (http://www.cpc.ncep.noaa.gov/products/analysis_monitoring/ensostuff/ensoyears.shtml). NAO is defined as a meridional dipole in atmospheric pressure with centres of action near the Azores and Iceland. A high NAO index is associated with stronger than average westerlies in the North Atlantic mid-latitudes, while the opposite is true for negative values of the index (Moore *et al.*, 2013). The NAO index is calculated by applying rotated principal component analysis, and here the values for the NAO index are taken from the Climate Prediction Center of NOAA (<http://www.cpc.ncep.noaa.gov/products/precip/CWlink/pna/nao.shtml>).

The IOD is a coupled ocean–atmosphere phenomenon, defined as the SST anomaly difference between the western equatorial Indian Ocean (50°–70°E, 10°S–10°N) and the south eastern equatorial Indian Ocean (90°–110°E, 10°S–0°N), referred to as Dipole Mode Index (DMI) (Saji *et al.*, 1999). Derived from Hadley Centre Sea Ice and Sea Surface Temperature (HadISST) dataset, there are monthly IOD from 1870 to present, maintained by the Japan Agency for Marine–Earth Science and Technology (JAMSTEC) (http://www.jamstec.go.jp/frsgc/research/d1/iod/iod/dipole_mode_index.html).

The PDO is most frequently referred to as a long-lived El-Niño like pattern of the Pacific climate variability. And the PDO is defined as the leading principal component of monthly SST anomalies in the North Pacific Ocean, poleward of 20°N (Zhang *et al.*, 1997). The values for the PDO index used here are taken from the Earth System Research Laboratory (ESRL) of NOAA (<http://www.esrl.noaa.gov/psd/data/correlation/pdo.data>), which contains data from 1948 to the present. It should be noted here that all of these indices used in the study are their annual average values for the period of 1959–2009, and the year is defined as from March to February in the next year.

3. Methodologies

3.1. Decompose the precipitation into spatial and the associated temporal patterns

Precipitation is characterized by nonlinearity and high dimensionality, in order to reduce the dimensionality of the system to a few modes, the rotated empirical orthogonal

functions (rotated EOFs) have been used to decompose the seasonal precipitation of all of the rain gauge stations into spatial patterns and the associated temporal patterns, respectively in the Yangtze River basin. In this study, seasonal precipitation was normalized as the precipitation anomaly values (the anomalous seasonal precipitation divided by the corresponding average seasonal aggregate precipitation for all years).

3.1.1. Empirical orthogonal functions

EOFs have been used in atmospheric science since the late 1940s by Obukhov (1947), Fukuoka (1951) and Lorenz (1956). Since then EOFs have become popular analysis tools and are widely used in climate research. The original aim of EOFs (Obukhov, 1947; Fukuoka, 1951; Lorenz, 1956) was to achieve a decomposition of a continuous space-time field $X(t, s)$, where t and s denote, respectively, the time and spatial position, as (Hannachi *et al.*, 2007):

$$X(t, s) = \sum_{k=1}^M c_k(t) u_k(s) \quad (1)$$

where M is the number of modes contained in the field, using an optimal set of basic functions of space $u_k(s)$ and expansion functions of time $c_k(t)$. In practice the EOF technique aims at finding a new set of variables that capture most of the observed variance from the data through linear combinations of the original variables. As EOFs have been extensively studied in the literature, the formulation and computation of EOFs are not introduced here and detailed analysis can be referred to these references and the references therein (Hannachi *et al.*, 2007; Navarra and Simoncini, 2010).

Generally, spatial orthogonality and temporal uncorrelation of the spatial and temporal patterns, respectively impose limits on physical interpretability of EOF patterns, as physical processes are not independent (Hannachi *et al.*, 2007). For the sake of interpretation, various methods have been proposed, and the rotation of EOFs is perhaps the most used method due in part to its simplicity.

3.1.2. Rotated EOFs

Rotation of the EOF patterns can systematically alter the structures of EOFs. Constraining the rotation to maximize a simplicity criterion, the rotated EOF patterns can be made simple. Given a $p \times m$ matrix $U_m = (u_1, u_2, \dots, u_m)$ of the leading m EOFs, the rotation is formally achieved by seeking an $m \times m$ rotation matrix R to construct the rotated EOFs B according to (Hannachi *et al.*, 2007):

$$B = U_m R \quad (2)$$

The criterion for choosing the rotation matrix R is what constitutes the rotation algorithm or the simplicity criterion, and is expressed by the maximization problem:

$$\max f(U_m R) \quad (3)$$

over a specified subset or class of $m \times m$ square rotation matrices R . The functional $f(\cdot)$ represents the rotation

criterion. There are various rotation criteria, and broadly speaking there are two large families of rotation: orthogonal and oblique rotations. However, as the orthogonal rotation is more efficient and computationally than oblique rotation due to the matrix inversion in the latter (Hannachi *et al.*, 2007), the orthogonal rotation has been used in this study.

In orthogonal rotation the rotation matrix R in Equation (2) is chosen to be orthogonal, and the problem is to solve Equation (3) subject to the condition (Hannachi *et al.*, 2007):

$$RR^T = R^T R = I_m \quad (4)$$

where I_m is the $m \times m$ identity matrix.

The most well-known and used rotation algorithm is the VARIMAX criterion (Hannachi *et al.*, 2007; Navarra and Simoncini, 2010). Let us designate by b_{ij} , $i = 1, \dots, p$, and $j = 1, \dots, m$, the elements of the rotated EOFs matrix B in Equation (2), then the VARIMAX orthogonal rotation maximizes a simplicity criterion according to (Hannachi *et al.*, 2007):

$$\max \left(f(B) = \sum_{k=1}^m \left[p \sum_{j=1}^p b_{jk}^4 - \left(\sum_{j=1}^p b_{jk}^2 \right)^2 \right] \right) \quad (5)$$

where m is the number of EOFs chosen for rotation. In some cases, to reduce the bias toward the first EOF with the largest eigenvalue, the loadings of the rotated EOFs B are weighted by the communalities of different variables. The communalities h_j^2 , $j = 1, \dots, p$, are directly

proportional to the sum of squares, $\sum_{k=1}^m u_{jk}^2$, of the loadings

for a particular variable. Hence if $C = \text{Diag}(U_m U_m^T)^{-1/2}$, then in the weighted or normalized VARIMAX, matrix B as used in Equation (5), is simply replaced by BC (Hannachi *et al.*, 2007).

In this study, the normalized VARIMAX has been used. However, to apply rotation, one has to determine the number of EOFs that need to be rotated. Technically, all the EOFs can be rotated, but this defeats the objective of dimension reduction and is likely to lead to trivial solutions. In addition, the EOFs scaled by the corresponding singular values seem to be better off with orthogonal rotation, to be robust with regard to changes in the number m of EOFs. However, such rotation of scaled EOFs excludes the contribution to the leading modes from low-ranked EOFs, with small eigenvalues, but which can be physically relevant (Hannachi *et al.*, 2006). In this article a series of number of unscaled EOFs were chosen to be rotated, and the well separated rotated EOFs were selected by using the screen graph to determine a sharp change in the convergence to zero of the explained variance (Navarra and Simoncini, 2010, Section 4.5.2) from the spectrum of the explained variance of the leading rotated EOFs, with a series of number of EOFs to be rotated. It should be noted here that in this manuscript we use the terminology REOFs and PCs for the spatial and temporal patterns of rotated EOFs, respectively.

3.2. Teleconnections between the climate indices and seasonal precipitation

The possible influences of ENSO, NAO, IOD and PDO on the seasonal precipitation regimes have been analysed by using the Pearson correlation coefficients between the ENSO, NAO, IOD and PDO and the temporal patterns of the rotated EOFs. As the climate indices may influence the precipitation in the Yangtze River basin of the year and the next year, and then the influences of the climate indices with 0- and 1-year ahead on the seasonal precipitation have been studied. Besides, to remove the influences of trend, the temporal patterns of rotated EOFs have been detrended (simple linear trend in mean has been removed by subtracting a least-squares-fit straight line).

3.3. Changes for the seasonal occurrence and intensity of wet days linked to the climate indices

To discover which of these climatic indices such as ENSO, NAO, IOD and PDO is the dominant mechanism driving seasonal precipitation changes, changes of the seasonal occurrence and intensity of wet days linked to the ENSO, NAO, IOD and PDO indices have been investigated in the study. Meanwhile, in the study, the changes of the seasonal occurrence and intensity of wet days linked to the climate index are denoted by the coefficient of the climate index, which represent the changes of seasonal occurrence and intensity of wet days corresponding to an increase of a unit of each climate index.

The coefficient of climate indices is calculated based on the linear regressions of the seasonal number and intensity of wet days on the climate index for all of the rain gauge stations in the Yangtze River basin. To remove the influences of trend, the number and intensity of wet days in each season have also been detrended.

4. Results and brief discussions

4.1. The selection of rotated EOFs mode for seasonal precipitation regimes

In the study, the first five modes of rotated EOFs have been selected to represent the seasonal precipitation regimes, almost explaining 50% of the total variance. The spectrum of the explained variance of the leading rotated EOFs for seasonal precipitation anomaly values, with 5, 8, 11 and 14 EOFs to be rotated has been illustrated in Figure 2. For spring precipitation, for the 11 EOFs to be rotated, there are sharp changes in the convergence to zero around the fifth and sixth leading rotated EOFs, and then the leading five rotated EOFs with 11 EOFs to be rotated have been retained, explaining, respectively, 17.6, 13.0, 11.1, 6.1 and 4.8% of the total variance. And for summer, there is a sharp change in the curve slope around the fifth and sixth leading rotated EOFs for the 14 EOFs to be rotated, and then the first five leading rotated EOFs with 14 EOFs to be rotated have been retained, explaining 16.3, 7.7, 7.5, 6.5 and 6% of the total variance. Similarly, for the autumn precipitation, the leading five rotated EOFs with 14 EOFs

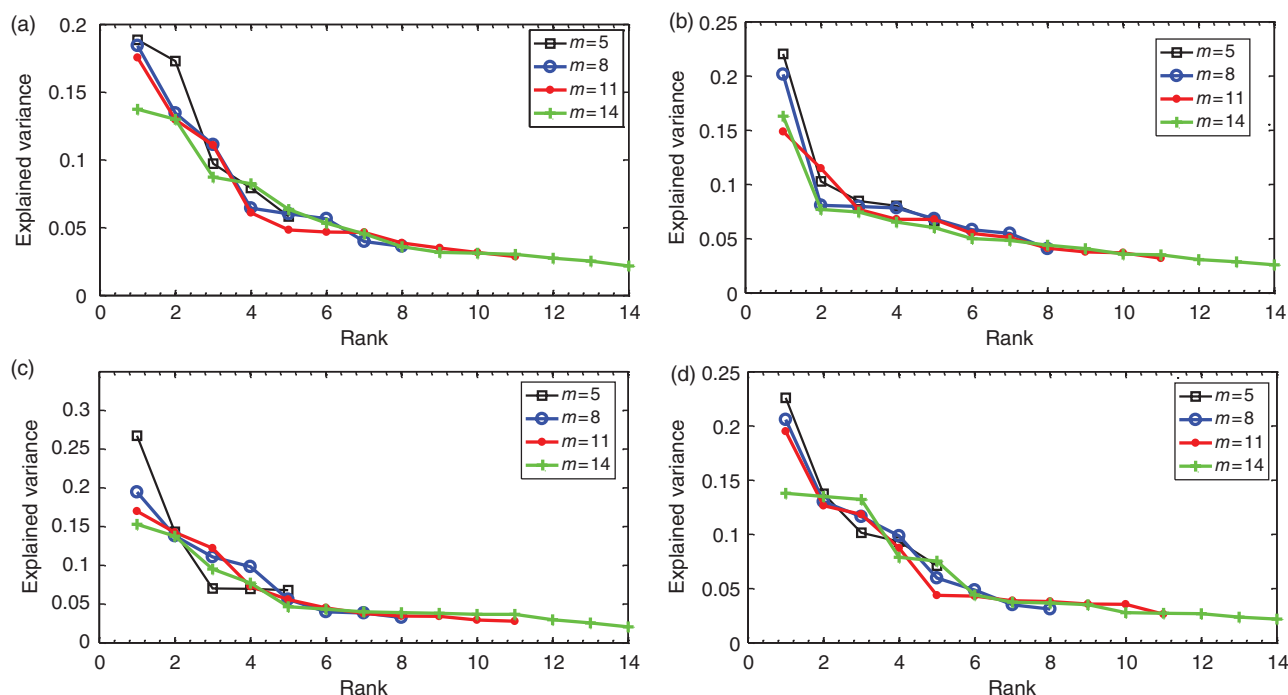


Figure 2. Spectrum of the explained variance of the leading REOFs of seasonal precipitation anomaly values, where m is the number of EOFs to be rotated, and (a), (b), (c) and (d) denote the spring, summer, autumn and winter, respectively.

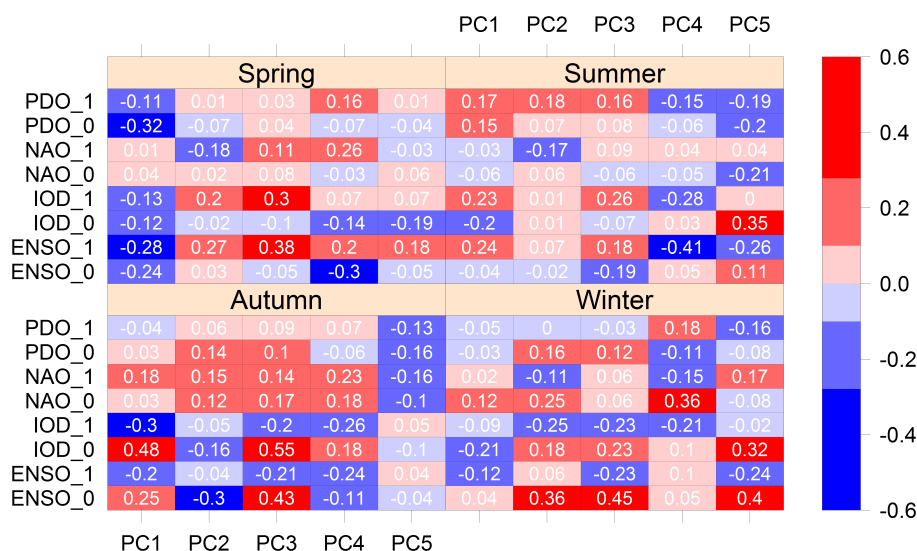


Figure 3. The Pearson correlation coefficient between the temporal pattern of the rotated EOFs and the climate indices, where the climate index with a suffix of '_0' and '_1' denote the climate index with 0- and 1-year ahead, respectively and the number with the dark colour indicates that it is significant at the 95% confidence.

to be rotated have been retained, explaining, respectively, 15.3, 13.8, 9.5, 7.7 and 4.7% of the total variance; and the leading five rotated EOFs with 11 EOFs to be rotated have been selected for the winter precipitation, explaining, respectively, 19.5, 12.7, 11.8, 8.7 and 4.4% of the total variance.

4.2. The relationship between the climate indices and the temporal patterns of rotated EOFs

The Pearson correlation coefficients between the temporal patterns of rotated EOFs and the climate indices have been

calculated in the study, shown in Figure 3. It can be seen from Figure 3 that for spring precipitation, the first mode of rotated EOFs is negatively correlated with the PDO at the same year, the third mode is positively correlated with the ENSO a year earlier and the fourth mode is negatively correlated with the ENSO at the same year. For summer precipitation, only the fourth and fifth mode of rotated EOFs are significantly influenced by the climate indices, and the fourth mode is negatively correlated with the ENSO a year earlier, the fifth mode is positively correlated with the IOD at the same year.

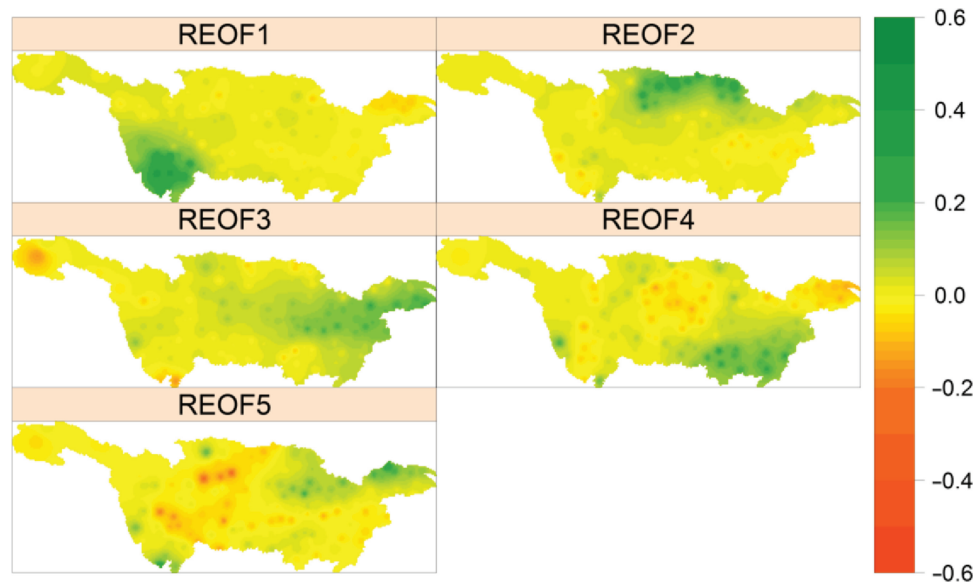


Figure 4. The spatial patterns of the leading five rotated EOFs for the spring precipitation anomalies.

Besides, for autumn precipitation, the leading three modes of rotated EOFs are influenced by the climate indices, and the first and third mode is positively correlated with the IOD at the same year, the second mode is negatively correlated with the ENSO at the same year. Also it can be seen from Figure 3 that the winter precipitation are mainly influenced by the ENSO and the NAO at the same year, and the second, third and fifth mode of rotated EOFs are positively correlated with the ENSO at the same year, the fourth mode is positively correlated with the NAO at the same year.

4.3. The spatial patterns of seasonal precipitation regimes influenced by the climate indices

The spatial patterns of the leading five rotated EOFs for the spring precipitation anomalies have been illustrated in

Figure 4, and as previously introduced in the Section 4.2 that the first, third and fourth mode of rotated EOFs for the spring precipitation have been influenced by the climate indices, then it can be seen from Figure 4 that the negative PDO event at the same year tends to increase the spring precipitation in the southwestern part of the Yangtze River basin (Figure 4, REOF1), and this is consistent with the results of Chan and Zhou (2005) that the early summer (May to June) monsoon rainfall over South China is related to PDO, with more dry (wet) monsoon years during the periods of high (low) PDO index. Besides, the positive ENSO event a year earlier tends to increase the spring precipitation in the east part of the Yangtze River basin (Figure 4, REOF3) and the positive ENSO event at the same year tends to decrease the spring precipitation in the southeastern part of the Yangtze River basin (Figure 4,

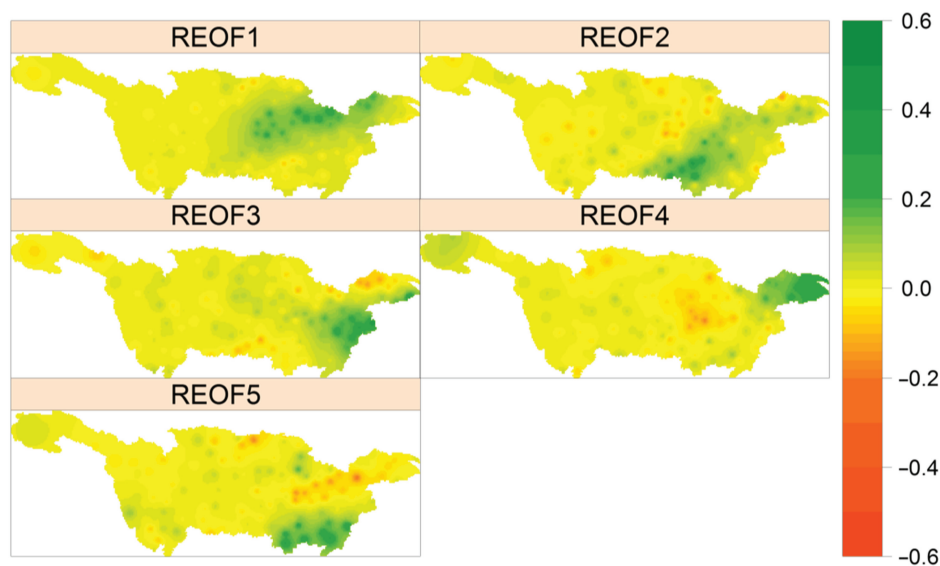


Figure 5. The spatial patterns of the leading five rotated EOFs for the summer precipitation anomalies.

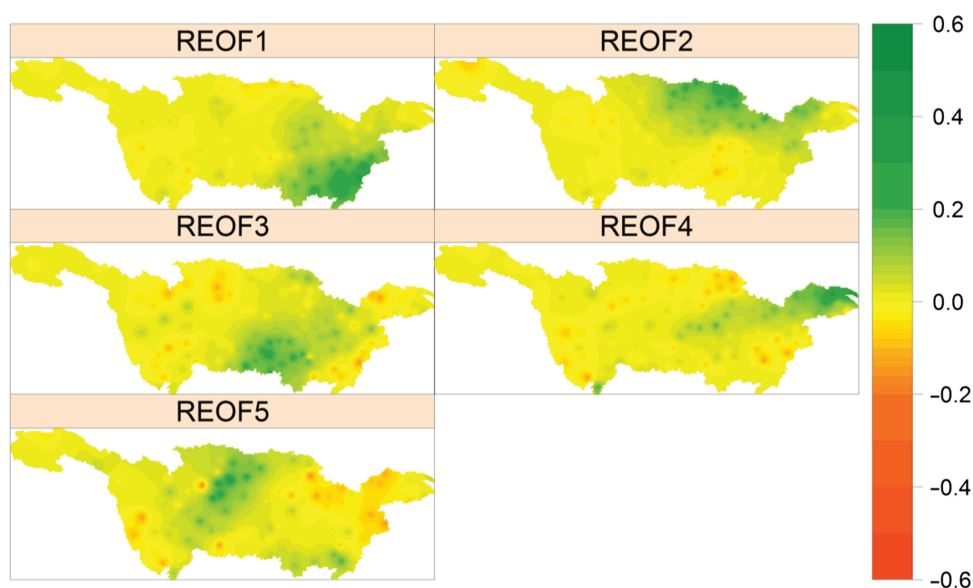


Figure 6. The spatial patterns of the leading five rotated EOFs for the autumn precipitation anomalies.

REOF4). And these may be caused by an anomalous lower-tropospheric anticyclone (cyclone) located in the western North Pacific during the ENSO event (Wang *et al.*, 2000; Chen *et al.*, 2013). The western North Pacific wind anomalies develop rapidly in the late fall of the year when a strong warm or cold event matures, and persist until the following spring or early summer, causing anomalously wet (dry) conditions along the East Asian polar front stretching from southern China northeastward to the east of Japan (Wang *et al.*, 2000).

For summer precipitation, the spatial patterns of the leading five rotated EOFs have been illustrated in Figure 5, and also as previously described in the Section 4.2 that the fourth and fifth mode of rotated EOFs for the summer precipitation have been influenced by the climate indices, then it can be seen from Figure 5 that the negative ENSO event a year earlier tends to increase the summer precipitation in the lower Yangtze River basin (Figure 5, REOF4) and the positive IOD event at the same year tends to increase the summer precipitation in the southeastern part of the Yangtze River basin (Figure 5, REOF5), which is consistent with the results of Liu *et al.* (2009) that summer rainfall will be more than normal in South China (centred in Hunan province) in the years of positive IOD occurring independently. Yuan *et al.* (2008) found that in the summer posterior to an IOD year, a positive (negative) IOD tends to induce an enhanced (a weakened) 500-hPa western Pacific subtropical high with a westward-advancing (an eastward-retreating) high ridge, and these will increase (decrease) rainfall in the upper and middle reaches of the Yangtze River Valley, China. And these may be the physical mechanisms behind the changes of summer precipitation caused by IOD (Yuan *et al.*, 2008).

Besides, the spatial patterns of the leading five rotated EOFs for autumn precipitation have been illustrated in Figure 6. As previously introduced in the Section 4.2 that the leading three modes of rotated EOFs for the autumn

precipitation have been influenced by the climate indices, it can be seen from Figure 6 that the positive IOD event at the same year tends to increase the autumn precipitation in the southeastern and centre part of Yangtze River basin (Figure 6, REOF1 and REOF3), and the influences of IOD on the maximum 1-day precipitation amount and maximum 5-day precipitation amount in the Poyang Lake basin (located in the middle Yangtze River basin) have also been found by Zhang *et al.* (2014). The negative ENSO event at the same year tends to increase the autumn precipitation in the south part of Yangtze River basin (Figure 6, REOF2).

For winter precipitation, the spatial patterns of the leading five rotated EOFs have been illustrated in Figure 7, and as previously introduced in the Section 4.2 that the winter precipitation are mainly influenced by the ENSO and NAO, then it can be seen from Figure 7 that the positive ENSO event at the same year tends to increase the winter precipitation in the southwestern and southeastern parts of Yangtze River basin (Figure 7, REOF2, REOF3 and REOF5), and the positive NAO event at the same year tends to increase the winter precipitation in the centre part of Yangtze River basin (Figure 7, REOF4), and which may be due to the anomaly easterly wind with maritime provenances during the positive NAO at the same year, referred to the Figure 13 of Zhang *et al.* (2014).

4.4. Changes for the seasonal occurrence and intensity of wet days linked to the climate indices

The coefficients of climate indices against the corresponding REOF value of all of the rain gauge stations for spring precipitation have been illustrated in Figure 8. It can be seen from Figure 8 that for the first mode of spring precipitation, for the stations with the linear regression significant at the 80% confidence, changes of the number of wet days linked to the PDO at the same year tends to be negative at the stations with positive REOF value while positive at

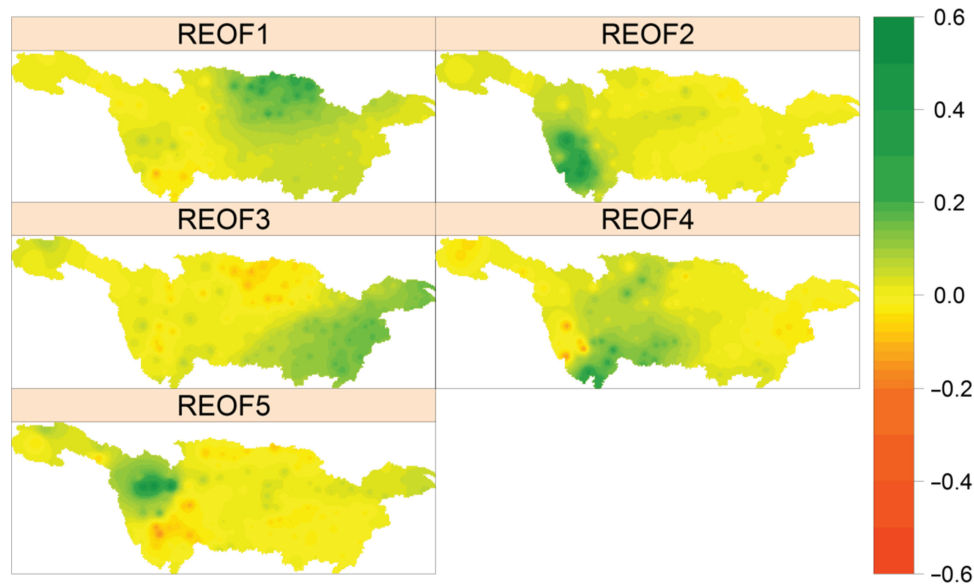


Figure 7. The spatial patterns of the leading five rotated EOFs for the winter precipitation anomalies.

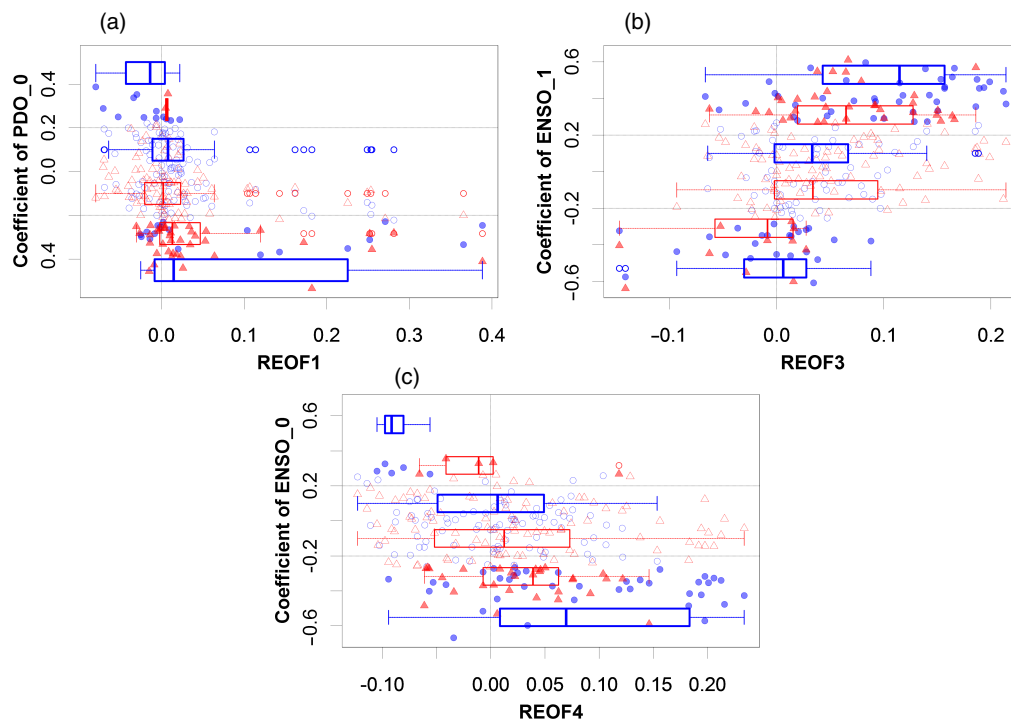


Figure 8. The coefficient of climate indices against the corresponding REOF value of all of the rain gauge stations in the Yangtze River basin for spring precipitation, where blue for changes of the number of wet days linked to the climate index, red for changes of the intensity of wet days linked to the climate index, and the solid points are significant at the 80% confidence level.

the stations with negative REOF value. These further verify the results described in Section 4.2 that the first mode of rotated EOFs for spring precipitation is negatively correlated with the PDO at the same year. Besides, there are no clear relationship between the changes of the intensity of wet days and the corresponding REOF value, then it can be concluded that the changes of spring precipitation linked to the PDO event at the same year are caused by the changes of occurrence of precipitation events. Similarly, for the third mode of spring precipitation, it can be

seen from Figure 8 that the changes of spring precipitation linked to the ENSO event a year earlier are mainly caused by changes of the intensity of precipitation events. However, for the fourth mode of spring precipitation, the changes of spring precipitation linked to the ENSO event at the same year are caused by changes of the occurrence and intensity of precipitation events simultaneously.

For summer precipitation, the coefficient of climate indices against the corresponding REOF value of all of the rain gauge stations have been illustrated in Figure 9. It can

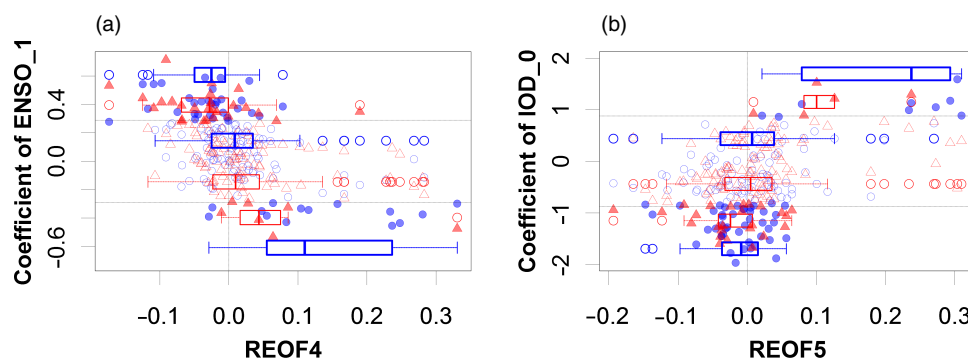


Figure 9. The coefficient of climate indices against the REOFs of all of the rain gauge stations in the Yangtze River basin for summer precipitation, where blue for changes of the number of wet days linked to the climate index, red for changes of the intensity of wet days linked to the climate index, and the solid points are significant at the 80% confidence level.

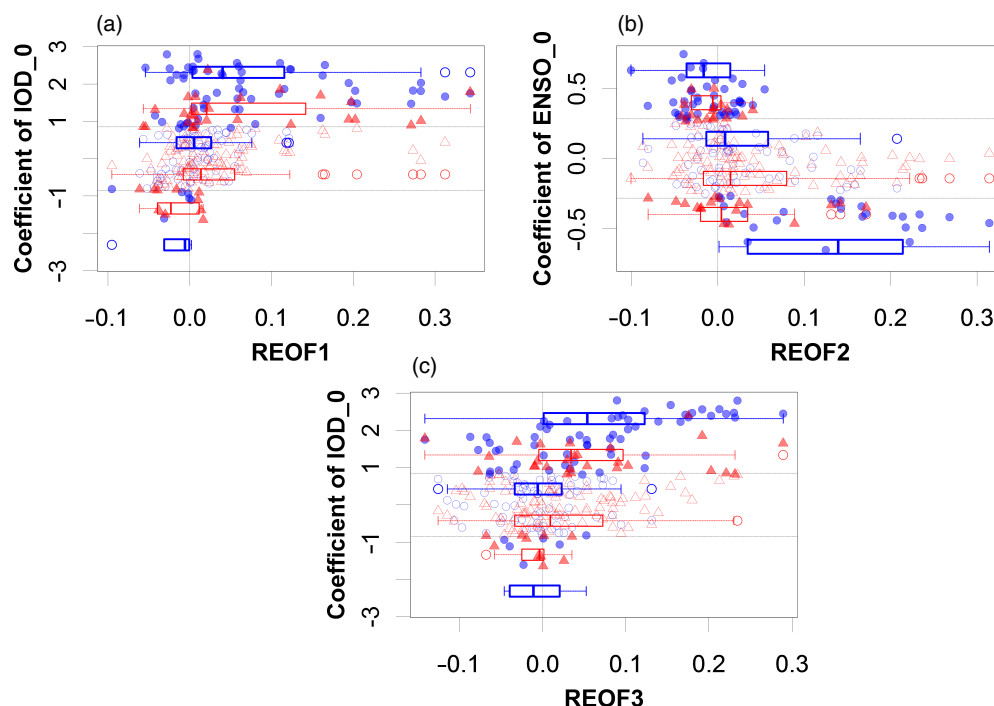


Figure 10. The coefficient of climate indices against the REOFs of all of the rain gauge stations in the Yangtze River basin for autumn precipitation, where blue for changes of the number of wet days linked to the climate index, red for changes of the intensity of wet days linked to the climate index, and the solid points are significant at the 80% confidence level.

be seen from Figure 9 that for the fourth mode of summer precipitation, changes of summer precipitation linked to the ENSO event a year earlier are caused by the changes of occurrence and intensity of precipitation events. And for the fifth mode of summer precipitation, changes of summer precipitation linked to the IOD event at the same year are also caused by changes of the occurrence and intensity of precipitation events.

Besides, the coefficients of climate indices against the corresponding REOF value of all of the rain gauge stations for autumn precipitation have been illustrated in Figure 10. It has been indicated previously that the first and third mode of autumn precipitation are influenced by the IOD event at the same year, however, it can be seen from Figure 10 that for the first mode of autumn precipitation, changes of autumn precipitation linked to

the IOD event at the same year are caused by the changes of occurrence and intensity of precipitation events simultaneously while for the third mode, changes of autumn precipitation are caused by the changes of intensity of precipitation events. Additionally, for the second mode of autumn precipitation, changes of autumn precipitation linked to the ENSO event at the same year are caused by changes of the occurrence of precipitation events.

Similarly, it has been described previously that the second, third and fifth mode of winter precipitation are influenced by the ENSO event at the same year, however, it can be seen from Figure 11 that for the second mode of winter precipitation, changes of winter precipitation linked to the ENSO event at the same year are caused by changes of the intensity of precipitation events while for the third mode, changes of winter precipitation are caused

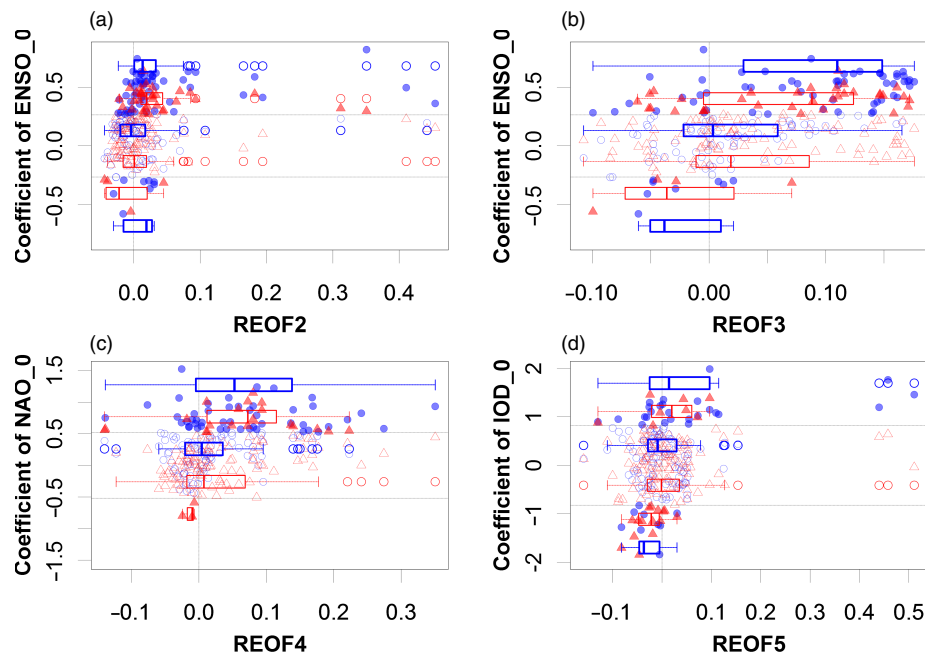


Figure 11. The coefficient of climate indices against the REOFs of all of the rain gauge stations in the Yangtze River basin for winter precipitation, where blue for changes of the number of wet days linked to the climate index, red for changes of the intensity of wet days linked to the climate index, and the solid points are significant at the 80% confidence level.

by changes of the occurrence and intensity of precipitation events simultaneously. For the fifth mode of winter precipitation, changes of the number and intensity of wet days linked to the ENSO event at the same year have no relationship with the corresponding REOF value (the figure has not been illustrated in the article). As the fifth mode of winter precipitation is also significantly positively correlated with IOD at the same year (shown in Figure 3), changes of winter precipitation linked to the IOD event at the same year have also been illustrated in Figure 11, it can be seen from Figure 11 that for the fifth mode of winter precipitation, changes of winter precipitation linked to the IOD event at the same year are caused by changes of the occurrence and intensity of precipitation events. Besides, for the fourth mode of winter precipitation, changes of winter precipitation linked to the NAO event at the same year are caused by changes of the intensity of precipitation events.

5. Conclusion

Using the rotated EOFs, the seasonal precipitation of all of the rain gauge stations have been decomposed into spatial patterns and the associated temporal patterns, respectively in the Yangtze River basin. Then based on the spatial patterns and the associated temporal patterns of seasonal precipitation, the possible influences of ENSO, NAO, IOD and PDO on seasonal precipitation in the Yangtze River basin have been analysed. Besides, to discover which is the dominant mechanism driving seasonal precipitation changes caused by ENSO, NAO, IOD and PDO, changes for the seasonal occurrence and intensity of wet days linked to the ENSO, NAO, IOD and PDO indices have also been investigated in the study. As a comprehensive of

all the above results, it can be concluded that for spring precipitation, the negative PDO event at the same year tends to increase the occurrence of precipitation events in the southwestern part of the Yangtze River basin. And the positive ENSO event a year earlier tends to increase the intensity of precipitation events in the east part of the Yangtze River basin, however, the positive ENSO event at the same year tends to decrease both the occurrence and intensity of precipitation events in the southeastern part of the Yangtze River basin.

For summer precipitation, both the occurrence and intensity of precipitation events have been influenced by the ENSO and IOD, the negative ENSO event a year earlier tends to increase the occurrence and intensity of precipitation events in the lower Yangtze River basin and the positive IOD event at the same year tends to increase the occurrence and intensity of precipitation events in the southeastern part of the Yangtze River basin.

For autumn precipitation, the positive IOD event at the same year tends to increase the autumn precipitation in the southeastern and centre part of Yangtze River basin, however, both the occurrence and intensity of precipitation events increase in the southeastern part of Yangtze River basin while only the intensity of precipitation events increases in the centre part of Yangtze River basin. The negative ENSO event at the same year tends to increase the occurrence of precipitation events in the south part of Yangtze River basin.

For winter, the precipitation in the southwestern and southeastern parts of Yangtze River basin have been influenced by the positive ENSO event at the same year, however, the positive ENSO event at the same year tends to only increase the intensity of precipitation events in

Table 1. A brief introduction of the coupled effects of climate indices on seasonal precipitation in the Yangtze River basin when at the same phase (positive or negative).

Season	The coupled effects of climate indices	
	Combination	Influences
Spring	PDO and ENSO	Strengthen the influences of ENSO on the precipitation occurrence in the lower Yangtze River basin and weaken the influences of PDO on the precipitation occurrence in the southwestern part of Yangtze River basin
Summer	ENSO and IOD	Weaken the influences of ENSO on both the precipitation occurrence and intensity in the lower Yangtze River basin
Autumn	IOD and ENSO	Weaken the influences of IOD on the precipitation occurrence in the centre part of Yangtze River basin
Winter	ENSO and NAO	Weaken the influences of ENSO on the precipitation intensity in the lower and southwestern part of Yangtze River basin
	ENSO and IOD	Weaken the influences of ENSO on the precipitation intensity in the southwestern part of Yangtze River basin
	NAO and IOD	Weaken the influences of IOD on the precipitation intensity in the west part of Yangtze River basin

southwestern parts of Yangtze River basin while both the occurrence and intensity of precipitation events have been increased in southeastern parts of Yangtze River basin. Besides, the positive IOD event at the same year tends to increase the occurrence and intensity of precipitation events in the west part of Yangtze River basin, and the positive NAO event at the same year tends to increase the intensity of precipitation events in the centre part of Yangtze River basin.

Furthermore, the coupled effects of ENSO, NAO, IOD and PDO on seasonal precipitation have also been concerned, and a brief introduction of the coupled effects of climate indices on seasonal precipitation in the Yangtze River basin has been summarized in Table 1. As the coupled effects of climate indices at the opposite phase are just the contrary for the coupled effects of climate indices at the same phase, only the coupled effects of climate indices at the same phase have been shown in the article. In summary, these will provide valuable information for improving the long-term forecasting of precipitation based on the relationship with ENSO, NAO, IOD and PDO, and will be helpful for planning and development of agricultural activities under influences of the natural climate variability.

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